

A Paradox in Machine Preference

softmax machines do not obey Luce’s Choice Axiom

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Abstract

Softmax output layers suggest, almost by construction, that large language models should obey Luce’s Choice Axiom: remove an option and the remaining probabilities renormalize. Using masked-token prompts (“my favorite Western state is [MASK]”) we find instead that the revealed preferences of BERT and RoBERTa are better matched by a Thurstonian contest model, in roughly 80% of cases. We extend the test to modern chat models via polysemous words, where Cekinmez et al. (2026) documented a gap between stated and generated sense distributions. The collapse replicates on Claude models, and under choice-set restriction the model re-collapses onto a runner-up sense rather than renormalizing—as the contest model predicts. We discuss Thurstonian layers and a principled diversity knob.

1 Introduction

Large language models can be interrogated cheaply and at scale, which makes the study of their revealed preferences attractive whether one regards them as incipient minds or as statistical mimics of our own predilections. Either way, a question from the psychology of choice arises immediately: when a model distributes probability over alternatives, what law governs how that distribution changes as the set of alternatives changes?

The workhorse answer in economics and psychology is Luce’s Choice Axiom [5], which asserts that the probability of selecting item i from a set S is proportional to a fixed utility $u(i)$:

$$P(i \mid S) = \frac{u(i)}{\sum_{j \in S} u(j)}. \quad (1)$$

Its signature is Independence of Irrelevant Alternatives: the ratio $P(A)/P(B)$ is unchanged by the presence or absence of a third option C .

There is a natural alignment between this assumption and the token probabilities of language models, which invariably emerge from a softmax:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}, \quad (2)$$

however complicated the transformer computation producing z_i may be. Delete item 1 from consideration—set $z_1 = -\infty$, a surgical procedure impossible in humans—and the remaining probabilities are the old ones renormalized by $1/(1 - p_1)$. In using softmax layers we are, in effect, heavily *suggesting* to the machine that it obey Luce’s Choice Axiom.

It doesn't. This paper demonstrates that language models, despite the suggestion built into their output layer, exhibit preference behavior better modeled in the Thurstonian framework of noisy contests [7]. Section 2 states the two competing models. Sections 3–4 test them on masked language models. Section 5 extends the test to modern chat models using polysemous words, connecting the paradox to the stated-versus-generated gap recently documented by Cekinmez, Wu, Marjeh and Griffiths [8]. Section 6 draws out the implications: a mechanism for generative “mode collapse,” a principled diversity knob, and a case for Thurstonian layers.

2 A choice of choice models

One might describe Equation (1) as an *urn model*: item k is represented by balls in an urn in proportion to $u(k)$, a choice is a random draw, and shrinking the choice set merely removes balls. The renormalization of probability under restriction then seems perfectly reasonable.

But perhaps preference is a horse race, not an urn. In the competing model each item i carries a location parameter a_i (a dislikability, say) and an associated performance $X_i \sim N(a_i, 1)$. Item i is chosen when its performance is the lowest draw. This is the model of Thurstone [7], and it makes different predictions about how choice probabilities respond when the choice set is altered: rather than removing a ball from the urn, we remove a horse from the race.

(As an aside, it *is* possible to engineer a contest that satisfies Luce's Choice Axiom, by taking $X_i \sim \text{Exp}(h_i)$ with common location and differing hazard rates. The comparison here can therefore be viewed as between two contest models, even if Luce's Choice Axiom is not usually presented that way.)

The practical obstacle to Thurstonian models has always been computation, not plausibility. A fast lattice algorithm for calibrating multi-entrant Thurstone models—recovering locations $\{a_i\}$ from winning probabilities and computing derived quantities for fields exceeding 100,000 entrants—was given in [1], and is what makes the experiments below routine.

3 Experimental design: masked language models

We probe the token probabilities of BERT [2] and RoBERTa [4], prompted to fill in blanks that invite choices among items of a category. Two designs are used.

In the first, the model answers an *original* question and a *qualified* question whose only difference is an adjective that shrinks the choice set:

1. *Original*: “My favorite state in the U.S. is [MASK] and I try to visit once a year.”
2. *Qualified*: “My favorite Western state in the U.S. is [MASK] and I try to visit once a year.”

In the second, we eliminate a single option explicitly, feeding the top answer of a first prompt back into a second:

1. *First*: “My favorite primary color is [MASK] because it is SOMETHING.”
2. *Second*: “My two favorite primary colors are [ANSWER] and [MASK] because they are SOMETHING.”

where SOMETHING ranges over many adjectives so that luck of phrasing is averaged out (the appendix of the working version lists roughly one hundred category/adjective combinations).

Both models make zero-parameter predictions of the restricted distribution from the unrestricted one. Luce’s prediction is renormalization, Equation (1); in the two-step design this is the simplest Plackett–Luce [6] or Harville [3] construction. The Thurstonian prediction calibrates locations $\{a_i\}$ to the unrestricted token probabilities via $\text{Prob}(X_i < X_k \ \forall k \neq i) = p_i$, removes the excluded contestant, and recomputes winning probabilities.

4 Results for masked language models

Table 1 shows the “favorite Western state” prompt for BERT. The Thurstonian column tracks the actual restricted probabilities more closely than Luce renormalization, which overweights the favorite (California) at the expense of the field. Table 2 repeats the exercise for “favorite girl’s baby name,” where the RMSE using Luce’s probabilities is roughly 50% higher.

State	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
California	10.89	29.87	25.38	20.17
Arizona	6.74	18.50	17.20	10.00
Texas	5.50	15.10	14.59	7.68
Colorado	3.18	8.74	9.38	7.81
Oregon	2.89	7.92	8.67	8.57
Oklahoma	1.97	5.41	6.37	6.64
Nevada	1.66	4.55	5.54	10.18
Montana	1.59	4.37	5.36	14.27
Idaho	1.05	2.87	3.82	4.99
Wyoming	0.98	2.69	3.62	9.69

Table 1: Predicted probabilities under Luce’s Choice Axiom and the Thurstonian model for the “favorite Western state” prompt (BertForMaskedLM). Both predictions use only the unqualified token probabilities; the Thurstonian model is closer to the actual qualified probabilities.

Name	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
Lily	1.56	20.35	16.18	14.52
Emily	1.15	15.06	13.31	16.29
Mia	0.86	11.25	11.02	13.18
Bella	0.85	11.12	10.93	8.84
Sarah	0.68	8.89	9.44	9.41
Rachel	0.59	7.66	8.56	7.15
Kate	0.59	7.63	8.54	8.44
Lauren	0.47	6.11	7.39	7.23
Brittany	0.46	6.04	7.33	7.55
Chloe	0.45	5.89	7.21	7.39

Table 2: The same comparison for “favorite girl’s baby name.” The RMSE using Luce’s probabilities is approximately 50% higher than the Thurstonian RMSE.

Across the full panel of qualified-question and single-item-elimination experiments, the Thursto-

nian model provides the more accurate prediction in approximately 80% of cases, with its advantage concentrated where the qualification is substantial and the entropy of the unrestricted distribution is low.

5 Polysemy as a preference probe

Cekinmez, Wu, Marjeh and Griffiths [8] recently turned an everyday property of language into a measurable distribution over meanings: give a model a bare polysemous word (“bolt”) with no disambiguating context, sample many outputs, and classify which sense each expresses. Across 32 text and image generation models they report a striking ladder of normalized sense entropies: image generation 0.10, text generation 0.25, human subjects 0.47, and—when models are asked to *predict* the distribution of senses—roughly 0.73. Models know the ambiguity; they do not express it. The finding sits within a growing literature on generative homogenization [9] and its sharpening under preference tuning [10]; their own ablation shows DPO alone cutting sense entropy from 0.25 to 0.18.

Read through the lens of Section 2, the ladder is what a contest model produces. Let the stated distribution proxy the latent locations $\{a_i\}$; let each generation be a contest among noisy sense performances. When noise is small relative to the spread of locations, winning probability is a violently convex function of location differences, and a mildly uneven prior collapses onto its mode. One noise parameter then indexes the entire ladder, and the “multimodal gap” becomes a noise gap rather than a knowledge gap. The rival explanation is a power tilt: renormalized p_i^γ , the Luce family with a temperature, which is mechanically what RLHF sharpening, low-temperature decoding, and classifier-free guidance perform. The two families are distinguished not by the collapse itself but by behavior under choice-set restriction—precisely the instrument of Section 3, and an experiment run by neither paper until now.

5.1 Replication on a chat model

We replicated the design at pilot scale on Claude models (October 2025 generation): 15 polysemous words from the stimulus set of [8], 30 independent sentence generations per word, senses classified by a second model against the closed inventory, plus a stated distribution elicited with their “predicting itself” framing. The collapse replicates almost exactly: mean normalized sense entropy 0.265 generated (their text-model figure: 0.25) against 0.906 stated. Eight of fifteen words collapsed to a single sense in all thirty samples.

Fitting the two one-parameter families from stated to generated distributions gives a statistical tie (Thurstone $\sigma = 0.49$, RMSE 0.293; power tilt $\gamma = 2.0$, RMSE 0.294; Thurstone the better fit on 10 of 15 words), and the reason both fit poorly is itself informative: for several words the generated mode is not even the stated argmax (“port” generates *harbor* in every sample while rating *connector* most probable). Stated reports are a biased window on the latent locations, not merely a flattened one—consistent with [8], whose stated distributions overestimate even human diversity. Calibration should therefore use revealed behavior, which is what the restriction design does.

5.2 Choice-set restriction

For the five words whose unrestricted distributions were non-degenerate, we re-sampled with the modal sense excluded by instruction and compared zero-parameter predictions of the restricted

distribution: Luce renormalization against Thurstonian contestant removal, both calibrated to the add-half smoothed unrestricted sense counts.

Word	Luce RMSE	Thurstonian RMSE	Restricted distribution
bolt	0.449	0.423	fastener .76, run .17, lock .07
seal	0.340	0.347	close .89, stamp .11
pitch	0.626	0.580	throw .06, tone .29, field .65
bat	0.750	0.709	animal 1.00, sports .00
iron	0.076	0.130	metal .93, press .07, golf .00
pooled	0.493	0.468	

Table 3: Restriction test on polysemous words (Claude Haiku 4.5, 30 samples per condition, modal sense excluded by instruction). The Thurstonian prediction is closer pooled and on three of five words.

The margin favors Thurstone, but the qualitative signature matters more than the margin: under restriction the model does not spread probability proportionally, as Luce requires. It *re-collapses* onto a new modal sense (seal $\rightarrow$ *close* at .89; iron $\rightarrow$ *metal* at .93). Remove the winning horse and the runner-up now wins nearly always—low-noise contest behavior, and the same low-entropy, heavy-qualification regime where the Thurstonian advantage concentrated in Section 4. One word (“bat”) flipped rank entirely under restriction, likely because excluding the *hit* sense also suppresses the overlapping *sports* sense; inventories with overlapping senses contaminate the test and neither family survives them.

6 Discussion

To the extent that LLMs mirror our own preferences, these results provide a rejection of Luce’s Choice Axiom in favor of the Thurstonian model. The popularity of the former may owe much to computational convenience, a consideration the algorithm of [1] removes.

The polysemy results suggest something further: that generative “mode collapse” is not a defect of knowledge but the lawful output of a low-noise contest run over intact locations. That reframing carries a practical remedy. If choice is a contest, diversity is restored not by a power tilt—raising the sampling temperature, which distorts the tails and does not renormalize the way behavior actually renormalizes—but by inverting the contest to recover locations and re-sampling at a chosen noise level. The calibration is exactly the fast Thurstone computation of [1], and it offers a principled diversity knob without retraining.

Leaning into the irony of softmax machines disobeying Luce’s Choice Axiom, the results also suggest that Thurstonian neural network layers might generalize better than softmax layers, the engineering challenge notwithstanding. Current architectures force hidden layers to fight the renormalization behavior their output layer implies; a contest-structured output layer would not need to.

Finally, this study is coarse-grained, and the pilot of Section 5 is small: fifteen words, one generation model, a single machine judge, sampling temperature not controlled. The full stimulus set and model panel of [8] would settle the one-parameter comparison with real power, and any model of machine decision-making that satisfies Luce’s Choice Axiom might, on this evidence, be viewed with suspicion—and readily improved.

Acknowledgments

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